Low-Velocity Impact Characteristics of 3D-Printed Poly-Lactic Acid Thermoplastic Processed by Fused Deposition Modeling

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ORIGINAL ARTICLE

Low-Velocity Impact Characteristics of 3D-Printed Poly-Lactic Acid Thermoplastic Processed by Fused Deposition Modeling

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Abstract The present research has focused on investigation into the low-velocity impact (LVI) behavior of 3D-printed poly-lactic acid thermoplastic material processed by fused deposition modeling. Four different infill density (ID) percentages, namely 40%, 60%, 80%, and 100%, were manufactured and subjected to the LVI test with 50 J energy input. The specimens were impacted using 12.7-mm hemispherical portion with conical head (ϕ 20 mm) striker which introduced two penetration stages. The impact characteristics in terms of velocity–time, force–time, and energy–time curves were investigated and reported. The penetration velocity limit, penetration energy, peak energy, and peak force were measured and discussed. The results revealed that the impact characteristics started to increase as the function of ID. The surface topography and fractured surfaces were examined using SEM. The fracture surfaces of 80% and 100% ID samples exhibited more viscous plastic deformation, indicating more energy absorption and load-carrying capacity.

Keywords Thermoplastic · Fused deposition modeling · Impact test · Microstructures · Damage evolution

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1 Introduction

The demand for producing environmentally unaffected, pollution-free, lightweight, and biocompatible products is increasing day-by-day in the world, which needs to avail the new additive manufacturing technology (AMT) [1–3]. AMT is a new one where the parts are manufactured layer-by-layer in a single machine [4, 5]. However, many tooling concepts, product design, process design, scheduling, routing, etc., are required in the conventional manufacturing method [6]. These can be eliminated in AMT except the product design. AMT includes fused deposition modeling (FDM), SLA, SLS, SLM, and LOM [7]. The major benefits of AMT are shorter product development time, effective material utilization, flexibility in both design and production of parts, less tooling cost, and being easy to produce intricate parts [8]. The detailed working of FDM and the schematic are explained elsewhere [9] and are included in the supplementary material (Figure S1). Poly-lactic acid (PLA) is an interesting, bio-based polymer and attract researchers which can be best suitable for tissue/bone engineering applications [10]. The PLA possesses considerable strength when compared to ABS [11]. Based on the literature [12–17] (see supplementary material), the mechanical properties in terms of tensile and compression tests on FDM were examined by various researchers. However, there is no literature to explain the LVI behavior of 3D-printed thermoplastic materials. Therefore, the main objectives of this research work are: to produce PLA parts using FDM technique by varying the ID and to examine the LVI behavior. The impact analyses on thermoplastic materials are also important when it is used as biodegradable and implant parts. These parts are also subjected to some impact during its applications, and hence, the present research work has been carried out. The FDM parts are susceptible to LVI which may cause significant damages, namely debonding between two layers, crack formation between the layers, and crack propagation, and breaking of any layer [14]. The ID has only varied in this work as it influences more on mechanical performance.

2 Material and Methods

The PLA filament of 1.75 mm diameter (BQ, Spain) was used; the physical and mechanical properties of PLA filament are illustrated in Table 1. The PLA samples were fabricated using WitBox 2 Ultimaker. The printer has a 0.4-mm nozzle diameter. The samples were printed with a dimension of 60 mm in diameter and 6 mm in thickness. The fixed parameters used in this work are presented in Table 2. The ID of 40%, 60%, 80%, and 100% were varied during FDM printing as ID influence is more on the dynamic mechanical behavior [17].

The LVI test was conducted using the CEAST 9350 impact tester. A 12.7-mm hemispherical portion with a conical head of 20 mm in diameter striking insert along with the tub holder was used. The geometry of the used striker was of the length of 40 mm with two different cross-sectional areas and made of tool steel hardened to 54 HRC. Two different cross-sectional areas of the striker offered two penetrating resistance during striking. This helped to predict the exact impact resistance of the materials. The total weight of 4.3 kg was used which included the striker and the carriage. A pneumatic system compressed the circular rings with an inner diameter of 40 mm which clamped the sample at 5.5 bar. The test was conducted at an ambient temperature of 25 °C. The impactor was dropped vertically downward with an incident velocity of 4.28 m/s and input energy of 50 J to ensure complete penetration of striker over the sample. The striker has fallen from a height of 942 mm. Three replicas of each ID were used during the impact test, and the average was used for investigation. The impact force was measured as a function of time using a load cell embedded on the impactor tool. The impactor incident velocity was measured through a photocell (optical detector). Table 3 illustrates the details of the sample size and the test parameters used in the present research work. The impactor velocity, displacement, and impact energy were calculated as stated by ASTM D7136 [18]:

Table 1 Physical and mechanical properties of as-received PLA filament

Properties	Values
Density	1.24 g/cm^3
Tensile strength	60 MPa
Flexural strength	108 MPa
Elongation	9%
Young's modulus	3100 MPa
Shore hardness, *D*	85 Sh D
Melting temperature	145–160 °C
Glass transition temperature	56–64 °C

Table 2 The fixed parameters used in fused deposition modeling using 3D printer

Variable	Selected	Variable	Selected
Material type	PLA	Layer height	200 µm
Filament diameter	1.75 mm	Platform adhesion type	Brim
Printing temperature	200 °C	Nozzle diameter	0.4 mm
Printing speed	20 mm/s	Air flow cooling	100%
Infill pattern	Line		

Table 3 Details of the samples and impact test parameters used in CEAST 9350

Parameters	Values, unit
Sample infill density	40, 60, 80, 100 (%)
Designed impact energy	50 J
Falling height	942 mm
Type of insert—hemispherical	12.7 mm
Incident velocity	4.28 m/s
Environment condition	Ambient temperature at 25 °C
Striker load capacity	21 kN
Sampling rate	850 kHz

$$v(t) = v_{\mathrm{i}} + gt - \int_0^t \frac{F(t)}{m}\mathrm{d}t \tag{1}$$

$$D(t) = D_{\mathrm{i}} + v_{\mathrm{i}}t + \frac{gt^2}{2} - \int_0^t \left(\int_0^t \frac{F(t)}{m}\mathrm{d}t\right)\mathrm{d}t \tag{2}$$

$$E(t) = \frac{m[v_i^2 - v(t)^2]}{2} + mgD(t) \tag{3}$$

where t represents the time in ms, v_{i} represents the impactor incident velocity in m/s, g represents the gravity $\mathrm{m/s^2}$, $F(t)$ represents the measured force in N, m represents the total mass (impactor and tub holder) in kg, D_{i} represents the initial position in mm, and $D(t)$ represents the displacement in mm.

The topography of the printed surface was examined using FEG HRSEM (30 keV, resolution of 1.3 nm at 1 kV) to check the printing quality. The fractured samples after impact test were also investigated using the same FEG HRSEM with different magnifications. Before capturing the SEM images, the sputtering of gold coating was done with a thickness of 15 nm. A back-scattered electron (BSE) detector was used to capture the SEM images.

3 Results

3.1 Surface Topography Examination on FDM-Printed PLA Samples

The surface topography of printed PLA samples with different IDs (40, 60, 80, and 100%) is observed using FEG HRSEM. Figure 1 shows the BSE SEM image on the printed PLA surface which clearly illustrates the material ID and the gap between the layers (porosity). Figure 2 demonstrates the decrease in porosity with the increase in ID as per designed input given into the FDM machine. There is no surface crack or damages that appear in the printed surface which ensure the good quality of FDM printing. For instance, all the layers in 100% ID sample are contacted with each other (Fig. 1d, 0% porosity). The dimension of a single layer corresponds to the nozzle diameter and the designed layer thickness. The nozzle diameter and the printing layer thickness are selected as 0.4 mm and 0.2 mm, respectively [19]. Hence, the printed layer cross section is exhibited as an ellipse which can be seen from the inset of Fig. 2a.

3.2 Low-Velocity Impact Responses on FDM-Printed PLA Samples

Figure 2 shows the schematic of the force–displacement curve which has been drawn based on the impact response of hemispherical portion with a conical-headed striker on PLA samples. The various traveling points of striker through the specimen during striking are also explained clearly in Fig. 2. The material offers first resistance against the penetration of the hemispherical portion (stage 1) and then the second resistance against the penetration of the conical-headed portion of the striker (stage 2). The observed impact results in terms of peak force, peak energy, penetration energy, and penetration limit velocity are given in Table 4.

Figure 3 shows the velocity–time impact characteristics curves of all the three replicas in each ID of PLA parts. The results produced by the three replicas of each ID are close to each other which ensures the printing quality of the machine used. In general, during LVI, the downward motion of striker has usually represented the positive velocity, whereas the upward motion of striker represents the negative velocity due to rebound/bounce. The rebound/

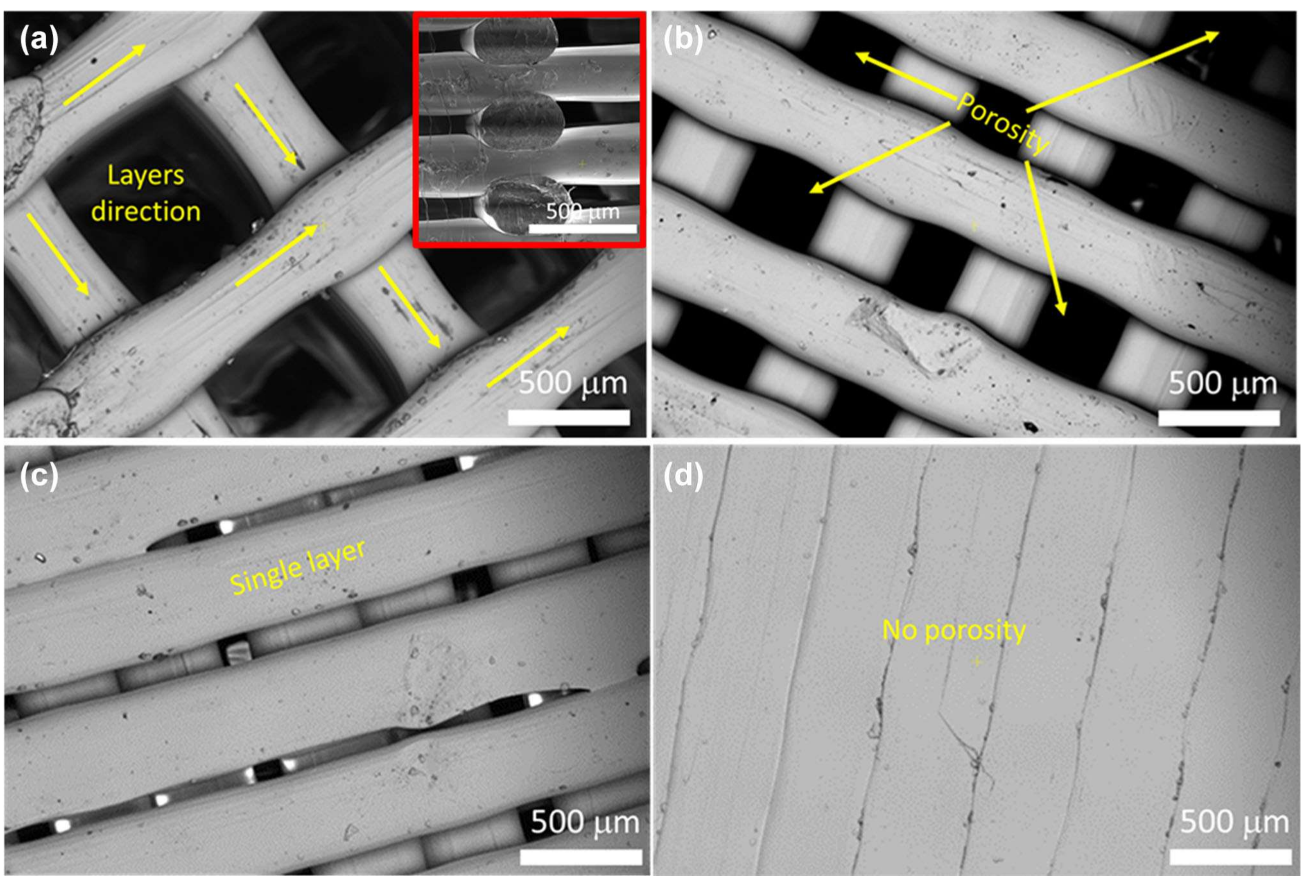

Fig. 1 SEM image of 3D-printed PLA samples at different infill density percentages: **a** 40%, **b** 60%, **c** 80%, and **d** 100%. Inset of **a** shows the cross section of 40% infill density sample

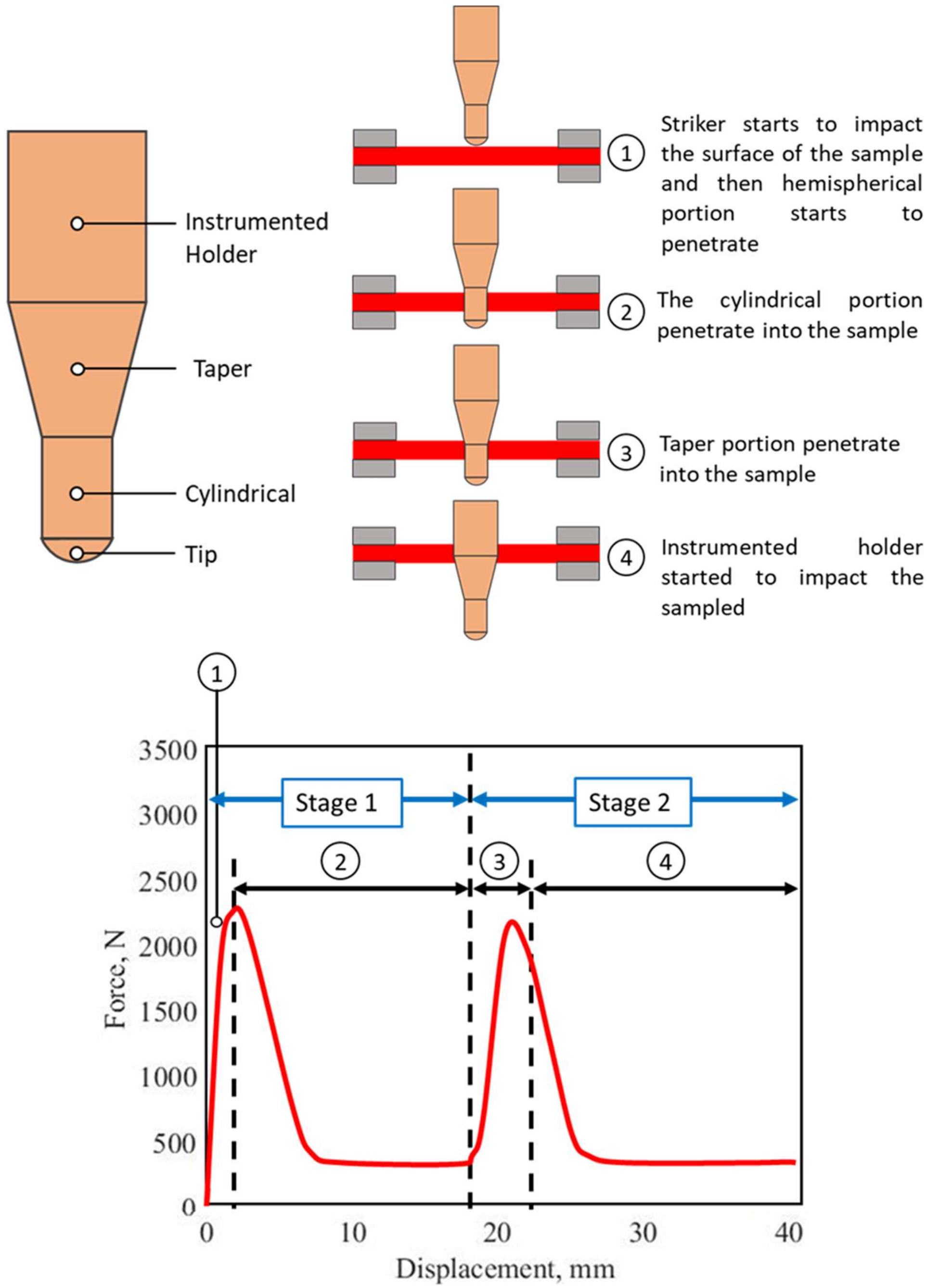

Fig. 2 Schematic of the force–displacement curve obtained during low-velocity impact test on PLA samples using hemispherical portion with conical-headed striker

bounce indicates that the striker does not penetrate into the sample for the given incident energy. The exhibited velocity of all samples (Fig. 3) produces positive value which means that the striker has penetrated completely. The velocity–time curves produces two different slopes during penetration which corresponds to the penetration of the hemispherical portion of the striker (stage 1) and the conical-headed portion (stage 2). The penetration limit in each stage can be calculated from the difference of striker velocity between the incident and residual velocities. The deceleration of striker starts to decrease significantly with the increase in ID in both the stages due to the increase in the stiffness of the materials. The penetration limit velocity increases with the function of ID due to increase in stiffness (Table 4). For instance, the average penetration limit velocity of 40% ID is 0.097 m/s and 0.273 m/s for stage 1 and stage 2, respectively. However, the average measured penetration limit velocity of 100% ID is 0.473 m/s and 1.067 m/s for stage 1 and stage 2, respectively. This is around 4.84 times (stage 1) and 3.91 times (stage 2) more in 100% ID samples when compared to 40% ID. The increase in ID increases the stiffness of the PLA samples

Table 4 Observed impact results of FDMed PLA samples

Sample detail	Peak force, N		Peak energy, J		Penetration energy, J		Penetration limit, m/s	
	Stage 1	Stage 2	Stage 1	Stage 2	Stage 1	Stage 2	Stage 1	Stage 2
40% ID								
Trial-1	495	605	3.95	12.05	1.549	3.857	0.085	0.215
Trial-2	560	775	3.51	10.35	1.909	6.009	0.105	0.340
Trial-3	490	615	2.75	8.3	1.819	4.726	0.100	0.265
Mean	515	665	3.4	10.23	1.759	4.864	0.097	0.273
SD	39	95	0.61	1.87	0.187	1.082	0.010	0.063
SE	22	55	0.35	1.08	0.108	0.625	0.006	0.036
60% ID								
Trial-1	1000	1405	7.00	25	4.985	13.720	0.280	0.825
Trial-2	985	1095	6.60	21.90	3.595	13.602	0.200	0.817
Trial-3	1410	1300	7.10	23.85	4.119	14.016	0.230	0.845
Mean	1131	1267	6.90	23.58	4.233	13.779	0.237	0.829
SD	241	158	0.26	1.57	0.702	0.214	0.040	0.014
SE	139	91	0.15	0.91	0.405	0.123	0.023	0.010
80% ID								
Trial-1	2355	2000	11	27.25	8.175	13.497	0.470	0.810
Trial-2	1985	1845	10	25	7.433	11.752	0.425	0.695
Trial-3	2005	1730	9.8	23.85	7.101	10.898	0.405	0.640
Mean	2115	1858	10.27	25.36	7.570	12.049	0.433	0.715
SD	208	135	0.64	1.72	0.550	1.325	0.033	0.087
SE	120	78	0.37	0.99	0.317	0.765	0.019	0.050
100% ID								
Trial-1	2750	1850	9.00	30.0	7.929	17.300	0.455	1.075
Trial-2	2450	2730	9.15	34.0	8.420	17.574	0.485	1.095
Trial-3	3400	2495	10.05	32.5	8.339	16.675	0.480	1.030
Mean	2867	2358	9.4	32.17	8.229	17.183	0.473	1.067
SD	485	455	0.56	2.02	0.263	0.461	0.016	0.033
SE	280	263	0.32	1.17	0.152	0.266	0.009	0.019

ID infill density, *SD* standard deviation, *SE* standard error

during impact loading. From these results, it can be understood that the penetration velocity limit depends on PLA ID and its deformation capability. To explain more, the penetration energy is also calculated from the energy conservation equation as described below [20]:

$$\text{Initial incident energy} = \text{penetration energy} + \text{residual energy} \tag{4}$$

$$\frac{1}{2}mv_{\mathrm{i}}^2 = E_{\mathrm{p}} + \frac{1}{2}mv_{\mathrm{r}}^2 \tag{5}$$

where m is the mass of the striking system in kg on the sample, v_{i} is the incident velocity in m/s, and v_{r} is the residual velocity in m/s.

The calculated penetration energy in each stage is also represented in Table 4. The observed penetration energy in each stage increases gradually with the function of ID. The average penetration energy in stage 1 is 1.76 J, 4,233 J, 7.57 J, and 8.29 J for 40, 60, 80 and 100% ID PLA samples, respectively. Similarly, the average penetration energy in stage 2 is 4.864 J, 13.779 J, 12.05 J, and 17.183 J for 40, 60, 80, and 100% ID PLA samples, respectively. These results can be attributed to decrease in porosity in the material due to which the material is able to absorb more amount of energy. In other words, the impact load-carrying capacity has improved with the function of ID increment. In addition, improved penetration energy of around 200–300% can be achieved in 100% ID PLA sample when compared to 40% ID. The observed penetration energy for 80% ID and 100% ID samples is near to each other. Hence, 80% ID percentage (20% porous) PLA sample can withstand more impact load, and simultaneously, the transferring of liquids/nutrition through pores can also be achieved. This combination of good strength and transferring of liquids/nutrition is an important

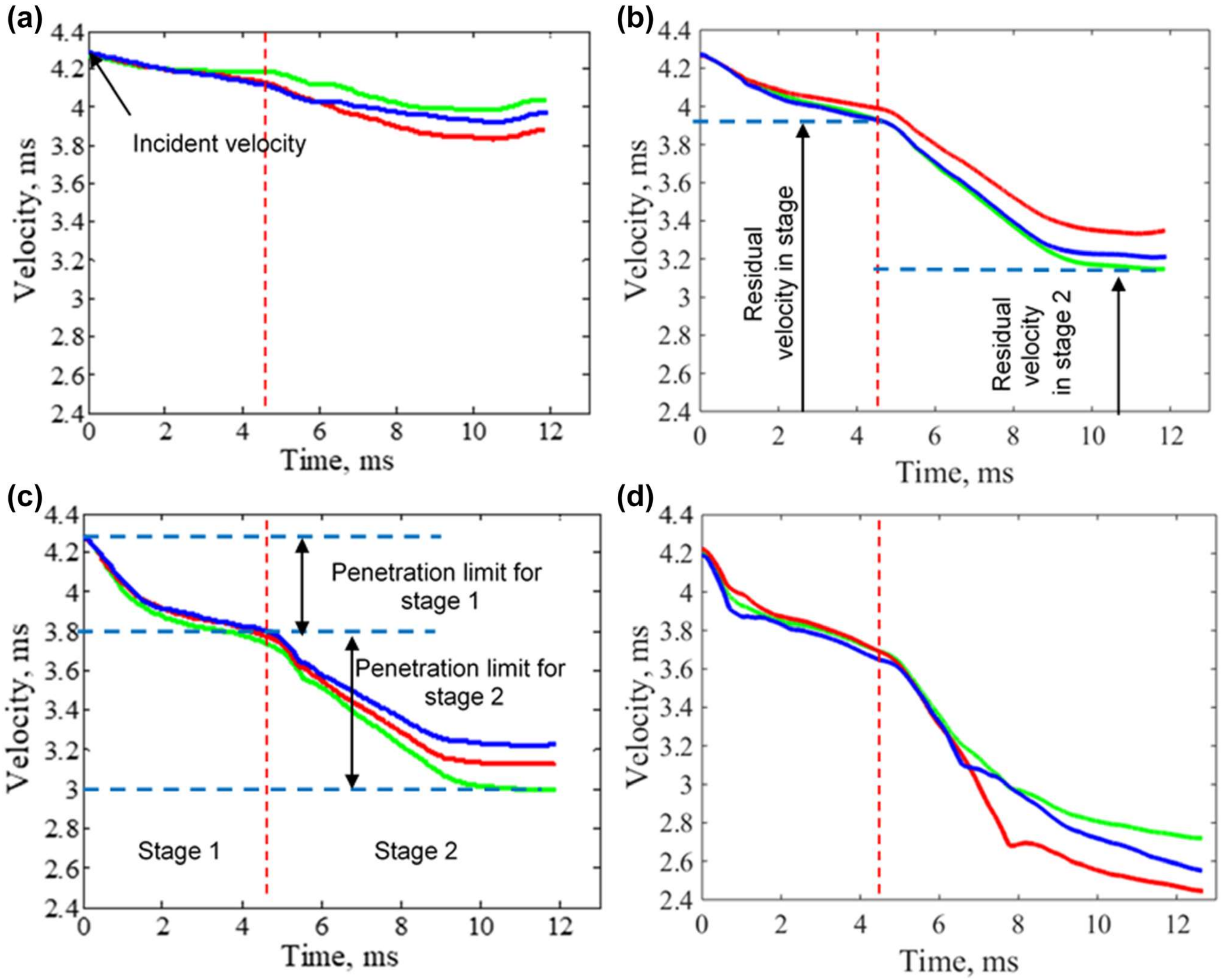

Fig.3 Impact velocity versus time characteristic curves for three replica samples in each infill density: **a** 40% ID; **b** 60% ID; **c** 80% ID; **d** 100% ID

characteristic in tissue/artificial bone engineering applications. Further, considerable absorbing energy capacity in addition to porosity is necessary for all medical/bio-medical components by which the stress shielding effect can be eliminated [21]. This stress shielding effect usually develops in the implant parts when the strength of the artificial bone/tissue does not match with the strength of real bones/tissues [22].

Figure 4 shows the impact force–time and energy–time responses. The observed behavior is similar to the penetration energy as explained in the earlier section. The results of force–time curves (Fig. 4a) reveal that the peaks force (in stage 1 and stage 2) starts to increase with the function of the ID. This is because of more layers are in contact at lower-porosity PLA sample, especially at 0% porosity (i.e., 100% infill density, Fig. 1d). In other words, the load-carrying capacity of high-dense PLA samples is enhanced when compared to the 40% ID PLA sample. From Fig. 4c and Table 4, the observed peak energy is consistent with the 80% and 100% ID PLA samples due to reduced porosity and increased bonding of each layer which is obtained from more ID. It is noted here that 80% ID sample exhibits around 26 J peak energy absorption, whereas the 100% ID sample produces 32 J peak energy absorption. These results demonstrate that both the samples possess good strength in which 80% ID PLA parts can be suggested for bio-medical/tissue parts as per the major requirement of porous structure. Based on the two peaks of force and energy (stage 1 and stage 2) obtained from each ID sample, a well-fitted line curves (Fig. 4b, d) along with the fitted equations are obtained by which the peak force and energy can be predicted exactly.

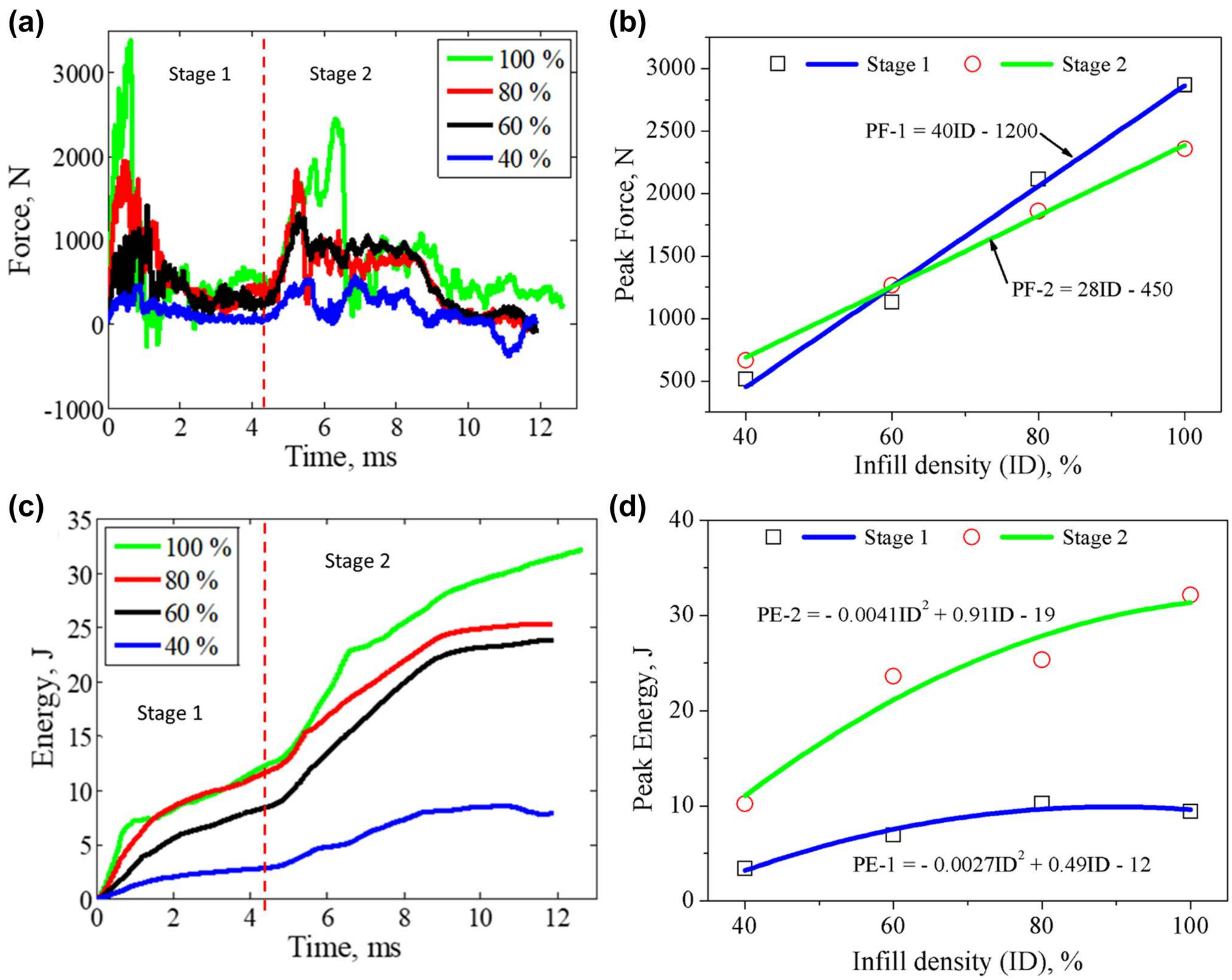

Fig. 4 Impact force and energy analyses: **a** comparison force profiles; **b** peak force as a function of the infill density percentage; **c** comparison of energy profiles; **d** peak energy as a function of the infill density percentage

3.3 SEM Fracture Surface Examination

The fractured surfaces have been examined using BSE of FEG HRSEM images (Fig. 5). From Fig. 5a, b of 40% ID PLA sample, the presence of more amount of porous region and less deformation region is observed. Here, layer-to-layer plastic deformation is very clear which indicates the coalescence of one layer with the other layer during deformation. These results ensure the effective bonding of one layer with the other layer during the FDM printing. Further, from Figs. 5c–i, more amount of plastic deformation and decrease in the porous region are observed due to the increase in ID. Especially, 80% and 100% ID PLA samples (Fig. 5e–i) produce more amount of severe viscous plastic deformation which indicates that these samples possess a high value of impact strength (Table 4), and hence, more amount of penetration energy and peak energy are produced (Table 4). The observed more viscous plastic deformation for 80% and 100% ID PLA samples suggest using it for bio-medical and even some structural parts. Further results are also presented in the supplementary material.

4 Conclusions

The LVI test on FDM PLA parts was executed by varying the ID. The printed samples were subjected to penetrate using hemispherical portion with a conical-headed striker. The results revealed that 80% and 100% ID PLA parts produced the absorbed peak energy of 1.2 to 2.5 times and 1.43 to 3 times higher than 40% and 60% ID PLA parts, respectively. Further, an improved penetration limit velocity was achieved in 80% and 100% ID PLA parts due to which it carried more impact load. This was due to more strength, reduction in porosity, and strong bonding between

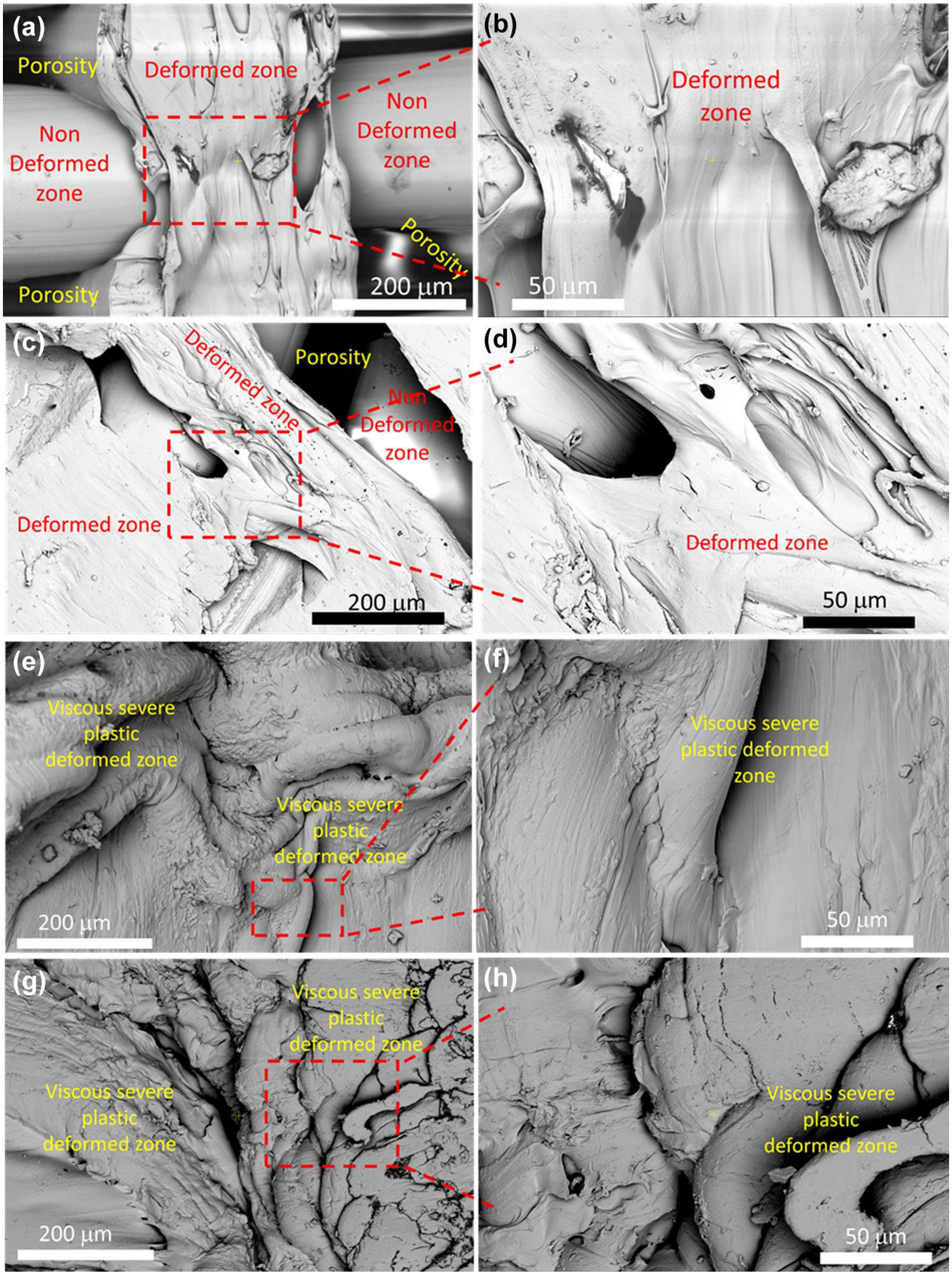

Fig. 5 FEI SEM low-velocity impact fracture surface images of FDM PLA samples of: left side **a** 40% ID, **c** 60% ID, **e** 80% ID, and **g** 100% ID; right side shows the corresponding magnified view

more layers. In addition, the improved energy absorption capacity of 80% and 100% ID PLA parts was evidenced with the macrostructure and microstructural evolutions. More viscous severe plastic deformation and more delamination pushdown damages occurred in 80% and 100% ID PLA parts due to considerable strength. In general, 80% and 100% ID PLA parts exhibited a good response on LVI.

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